Terraform-02

1). Watch the **Terraform-02** video

Using Providers configuration

```
main.tf > ...
}
resource "random_pet" "mypet" {
prefix = "MR"
separator = "."
length = "1"
}
```

- Now do terraform init it downloads dependencies

```
PS C:\Users\LENOVO\Desktop\Terraform-basic> terraform init
Initializing provider plugins found in the configuration...
- Finding latest version of hashicorp/random...
- Reusing previous version of hashicorp/local from the dependency lock file
- Installing hashicorp/random v3.9.0...
- Installed hashicorp/random v3.9.0 (signed by HashiCorp)
- Using previously-installed hashicorp/local v2.9.0

Initializing the backend...

Initializing provider plugins found in the state...
- Reusing previous version of hashicorp/local
- Using previously-installed hashicorp/local v2.9.0

Terraform has made some changes to the provider dependency selections recorded
in the .terraform.lock.hcl file. Review those changes and commit them to your
version control system if they represent changes you intended to make.

Terraform has been successfully initialized!
```

- Now Terraform plan it will destroy old data and add new one

```
PS C:\Users\LENOVO\Desktop\Terraform-basic> terraform plan
local_file.my_pets: Refreshing state... [id=24899fdea020fed9c33448b53811e17d15bb4f25]

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the
  + create
  - destroy

Terraform will perform the following actions:

  # local_file.my_pets will be destroyed
  # (because local_file.my_pets is not in configuration)
  - resource "local_file" "my_pets" {
      - content              = "i love pets" -> null
      - content_base64sha256 = "MosGGdd1+JiYF4ysx6GJenzof2jRKrkSPRV/oqnnaUY=" -> null
      - content_base64sha512 = "ew9wDoZ47WOyhSU2VZtKQs6GF8iJeW125t8oNbnNeySCBNgDzxPgaoSO8hihrWnN078rOk0wlTfbs416lYXcMA==
      - content_md5          = "85c1074c6f4e7a09d7fe0ec4bf55d15c" -> null
      - content_sha1         = "24899fdea020fed9c33448b53811e17d15bb4f25" -> null
      - content_sha256       = "328b0619d775f89898178cacc7a1897a7ce87f68d12ab9123d157fa2a9e76946" -> null
      - content_sha512       = "7b0f700e8678ed63b2852536559b4a42ce8617c889796d76e6df2835b9cd7b248204d803cf13e06a848ef218
```

```
  # random_pet.mypet will be created
  + resource "random_pet" "mypet" {
      + id        = (known after apply)
      + length    = 1
      + prefix    = "MR"
      + separator = "."
    }

Plan: 2 to add, 0 to change, 1 to destroy.
```

- Now Terraform apply Id create

```
PS C:\Users\LENOVO\Desktop\Terraform-basic> terraform apply
local_file.my_pets: Refreshing state... [id=24899fdea020fed9c33448b53811e17d15bb4f25]

Terraform used the selected providers to generate the following execution plan. Resource actio
  + create
  - destroy

Terraform will perform the following actions:

  # local_file.my_pets will be destroyed
  # (because local_file.my_pets is not in configuration)
  - resource "local_file" "my_pets" {
      - content              = "i love pets" -> null
      - content_base64sha256 = "MosGGdd1+JiYF4ysx6GJenzof2jRKrkSPRV/oqnnaUY=" -> null
      - content_base64sha512 = "ew9wDoZ47WOyhSU2VZtKQs6GF8iJeW125t8oNbnNeySCBNgDzxPgaoSO8hihrW
```

```
  # random_pet.mypet will be created
  + resource "random_pet" "mypet" {
      + id        = (known after apply)
      + length    = 1
      + prefix    = "MR"
      + separator = "."
    }

Plan: 2 to add, 0 to change, 1 to destroy.

Do you want to perform these actions?
  Terraform will perform the actions described above.
  Only 'yes' will be accepted to approve.

  Enter a value: yes
```

```
local_file.my_pets: Destroying... [id=24899fdea020fed9c33448b53811e17d15bb4f25]
random_pet.mypet: Creating...
local_file.my_pets: Destruction complete after 0s
local_file.pet: Creating...
random_pet.mypet: Creation complete after 0s [id=MR.deer]
local_file.pet: Creation complete after 0s [id=fefacccdae259f25533749abfb90e27558256459]

Apply complete! Resources: 2 added, 0 changed, 1 destroyed.
PS C:\Users\LENOVO\Desktop\Terraform-basic>
```

```
.terraform.lock.hcl
main.tf
pets.txt
terraform.tfstate
terraform.tfstate.backup
```

```
main.tf    pets.txt ×
pets.txt
i love pets
```

Now create an variable.tf configure it

```
variables.tf ×    var.filename
variables.tf > variable "content"
variable "filename" {
default = "pets.txt"
type = string
}

variable "content" {
  default = "I love cats"
}
```

```
main.tf > ...
resource "local_file" "pet" {
  filename = "pets.txt"
  content  = "i love pets"

  lifecycle {
    create_before_destroy = true
  }
}

```

```
PS C:\Users\LENOVO\Desktop\Terraform-basic> terraform apply
random_pet.petname: Refreshing state... [id=MR.lab]
local_file.my_pet: Refreshing state... [id=2560517838ec2e0d5f58a3963f91831616d0ad5d]

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
  + create
  - destroy

Terraform will perform the following actions:

  # local_file.my_pet will be destroyed
  # (because local_file.my_pet is not in configuration)
  - resource "local_file" "my_pet" {
```

```
  Enter a value: yes

local_file.my_pet: Destroying... [id=2560517838ec2e0d5f58a3963f91831616
random_pet.petname: Destroying... [id=MR.lab]
random_pet.petname: Destruction complete after 0s
local_file.pet: Creating...
local_file.my_pet: Destruction complete after 0s
local_file.pet: Creation complete after 0s [id=24899fdea020fed9c33448b5

Apply complete! Resources: 1 added, 0 changed, 2 destroyed.
```

- In main.tf added lifecycle

```
main.tf
main.tf > resource "local_file" "pet"
resource "local_file" "pet" {
  filename = "pets.txt"
  content  = "i love pets"

  lifecycle {
    ignore_changes = [
      content
    ]
  }
}
```

```
PS C:\Users\LENOVO\Desktop\Terraform-basic> terraform plan
random_pet.petname: Refreshing state... [id=MR.parrot]
local_file.my_pet: Refreshing state... [id=72cb499b8302a4cbaeb7262125680cc675ec9401]

Terraform used the selected providers to generate the following execution plan. Resou
  + create
  - destroy
```

```
  # random_pet.petname will be destroyed
  # (because random_pet.petname is not in configuration)
  - resource "random_pet" "petname" {
      - id        = "MR.parrot" -> null
      - length    = 1 -> null
      - prefix    = "MR" -> null
      - separator = "." -> null
    }

Plan: 1 to add, 0 to change, 1 to destroy.
```

```
PS C:\Users\LENOVO\Desktop\Terraform-basic> terraform apply
local_file.my_pet: Refreshing state... [id=72cb499b8302a4cbaeb7262125680cc675ec9401]
random_pet.petname: Refreshing state... [id=MR.parrot]

Terraform used the selected providers to generate the following execution plan. Resource
```

```
  # random_pet.petname will be destroyed
  # (because random_pet.petname is not in configuration)
  - resource "random_pet" "petname" {
      - id        = "MR.parrot" -> null
      - length    = 1 -> null
      - prefix    = "MR" -> null
      - separator = "." -> null
    }

Plan: 1 to add, 0 to change, 1 to destroy.
```

```
  Enter a value: yes

random_pet.petname: Destroying... [id=MR.parrot]
random_pet.petname: Destruction complete after 0s
local_file.pet: Creating...
local_file.pet: Creation complete after 0s [id=24899fdea020fed9c33448b5

Apply complete! Resources: 1 added, 0 changed, 1 destroyed.
```

2). Execute all the **templates shown in the video**.

Using two Providers in a same configuration

```
main.tf   variables.tf                      variables.tf   var.filename

main.tf > resource "local_file" "my_pet" > content      var.filename
resource "local_file" "my_pet" {                        var.content
filename = "var.filename"
content = "var.content"
}
resource "random_pet" "petname" {
prefix = "MR"
separator = "."
length = "1"
}
```

Now do Terraform apply

```
PS C:\Users\LENOVO\Desktop\Terraform-basic> terraform apply
local_file.pet: Refreshing state... [id=fefacccdae259f25533749abfb90e27558256459]
random_pet.mypet: Refreshing state... [id=MR.filly]

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated
with the following symbols:
  + create
  - destroy
```

- It will destroy old configuration and added new one

```
  # local_file.pet will be destroyed
  # (because local_file.pet is not in configuration)
  - resource "local_file" "pet" {
      - content              = "we love pets!" -> null
      - content_base64sha256 = "rsajQXVQs7YWi9t6jzWJaEoVMTLLjcmsutD0NSZ
      - content_base64sha512 = "CxNZ3FK+yyxcqVBkwBiYG2woxkuvPtXA0tDMnfn
      - content_md5          = "f08f69fe982581de9d6a2cf01fc6d5f8" -> nu
      - content_sha1         = "fefacccdae259f25533749abfb90e27558256459
      - content_sha256       = "aec6a3417550b3b6168bdb7a8f3589684a153132
      - content_sha512       = "0b1359dc52becb2c5ca95064c018981b6c28c64
" -> null
      - directory_permission = "0777" -> null
      - file_permission      = "0777" -> null
      - filename             = "/root/pets.txt" -> null
      - id                   = "fefacccdae259f25533749abfb90e27558256459
    }

  # random_pet.mypet will be destroyed
  # (because random_pet.mypet is not in configuration)
```

```
Plan: 2 to add, 0 to change, 2 to destroy.

Do you want to perform these actions?
  Terraform will perform the actions described above.
  Only 'yes' will be accepted to approve.

  Enter a value: yes

random_pet.mypet: Destroying... [id=MR.filly]
local_file.pet: Destroying... [id=fefacccdae259f25533749abfb9
random_pet.petname: Creating...
local_file.my_pet: Creating...
random_pet.mypet: Destruction complete after 0s
local_file.pet: Destruction complete after 0s
random_pet.petname: Creation complete after 0s [id=MR.lab]
local_file.my_pet: Creation complete after 0s [id=2560517838e

Apply complete! Resources: 2 added, 0 changed, 2 destroyed.
```

- Where we can see main.tf we mention what do in variable.tf configure

main.tf × | variables.tf

main.tf > resource "local_file" "my_pet"

```
resource "local_file" "my_pet" {
filename = "var.filename"
content = "var.content"
}
resource "random_pet" "petname" {
prefix = "MR"
separator = "."
length = "1"
}
```

variables.tf × | var.filename

variables.tf > variable "filename"

```
variable "filename" {


}
variable "content" {

}
```

- Now we can see var.content, var.filename both values enter

```
PS C:\Users\LENOVO\Desktop\Terraform-basic> terraform apply
var.content
  Enter a value: i hate cats

var.filename
  Enter a value: pets

random_pet.petname: Refreshing state... [id=MR.lab]
local_file.my_pet: Refreshing state... [id=72cb499b8302a4cbaeb7262125680cc675ec9401]
```

- Now it is replaced by old data configuration by terraform apply

```
 # local_file.my_pet must be replaced
-/+ resource "local_file" "my_pet" {
      ~ content              = "I love ABD" -> "var.content" # forces replacement
      ~ content_base64sha256 = "F/nIEzo9ESic6Upb8o6M/6LJw6atAThYNEi5yF44iGc=" -> (known after
      ~ content_base64sha512 = "Tm/V9KEo8XsSkpIOoQSCtCybYXhNwFe1MIcy1YvRoLY75LbFIDbJAtzF30q0Z
      ~ content_md5          = "3ead07d3e2fc0e4780f4b73ee1a490b5" -> (known after apply)
      ~ content_sha1         = "72cb499b8302a4cbaeb7262125680cc675ec9401" -> (known after app
```

```
Plan: 1 to add, 0 to change, 1 to destroy.

Do you want to perform these actions?
  Terraform will perform the actions described above.
  Only 'yes' will be accepted to approve.

  Enter a value: yes

local_file.my_pet: Destroying... [id=72cb499b8302a4cbaeb726212
local_file.my_pet: Destruction complete after 0s
local_file.my_pet: Creating...
local_file.my_pet: Creation complete after 0s [id=2560517838ec

Apply complete! Resources: 1 added, 0 changed, 1 destroyed.
```

- In variable.tf I given some data and it will create a file for variable

```
main.tf    variables.tf ×    pets.
variables.tf > variable "length"
variable "filename" {
default ="pets.txt"
}
variable "content" {
default ="I Like ABD!"
}
variable "prefix" {
default ="MR"
}
variable "separator" {
default ="."
}
variable "length" {
default ="1"
}
```

- Now do Terraform apply it will replaced and created a file for variable

```
PS C:\Users\LENOVO\Desktop\Terraform-basic> terraform apply
local_file.my_pet: Refreshing state... [id=2560517838ec2e0d5f58a3963f91831616d0ad5d]
random_pet.petname: Refreshing state... [id=MR.lab]

Terraform used the selected providers to generate the following execution plan. Resource actions a
-/+ destroy and then create replacement

Terraform will perform the following actions:

  # local_file.my_pet must be replaced
-/+ resource "local_file" "my_pet" {
      ~ content              = "var.content" -> "I love ABD" # forces replacement
      ~ content_base64sha256 = "8+TT03U9LszQJb/jSHweRYsj7ueiWTITMP0XjWQJJC0=" -> (known after appl
```

```
variables.tf    var.filename ×
var.filename
1    I love ABD
```

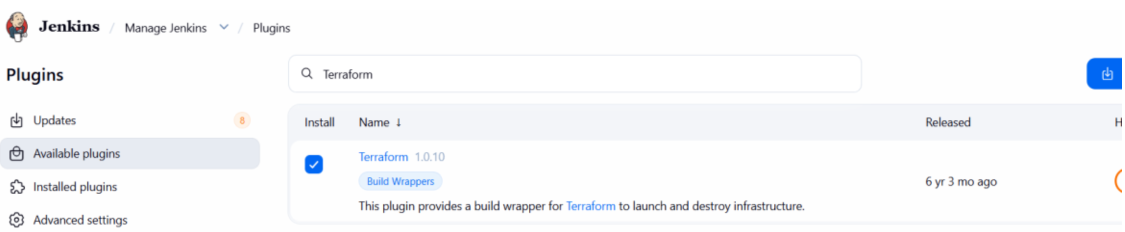
3). Integrate **Terraform in Jenkins** using the **Terraform plugin**.

- Open Jenkins server and check running or not and version also

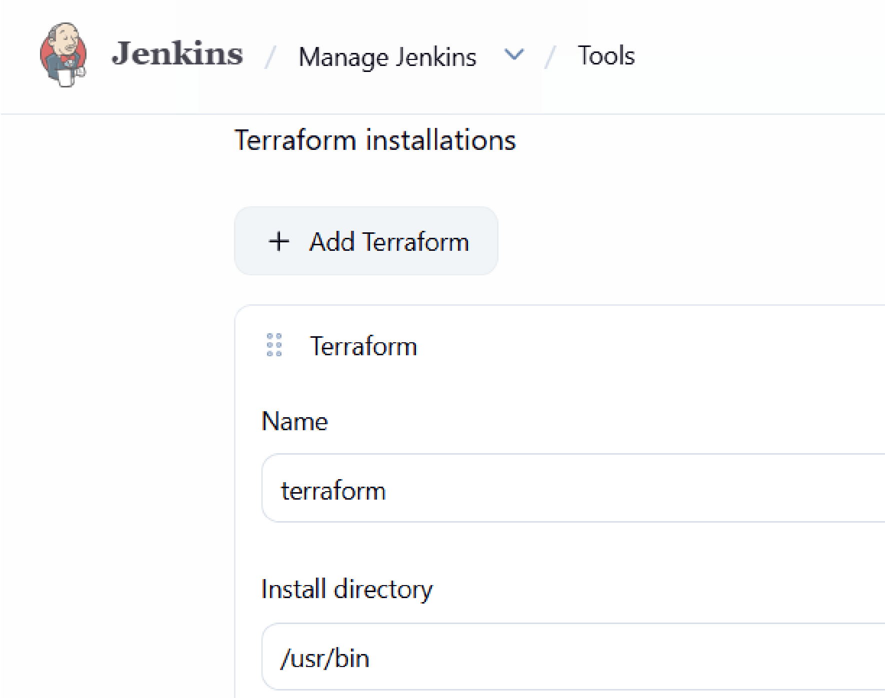
```
root@ip-172-31-22-87:~# systemctl start jenkins
root@ip-172-31-22-87:~# systemctl status jenkins
● jenkins.service - Jenkins Continuous Integration Server
     Loaded: loaded (/usr/lib/systemd/system/jenkins.service; enabled; preset: enabled)
     Active: active (running) since Tue 2026-06-09 17:22:57 UTC; 37s ago
   Main PID: 519 (java)
      Tasks: 45 (limit: 1013)
     Memory: 389.3M (peak: 390.6M)
        CPU: 22.347s
     CGroup: /system.slice/jenkins.service
             └─519 /usr/bin/java -Djava.awt.headless=true -jar /usr/share/java/jenkins.wa
```

```
root@ip-172-31-22-87:~# terraform -v
Terraform v1.15.5
on linux_amd64
root@ip-172-31-22-87:~#
```

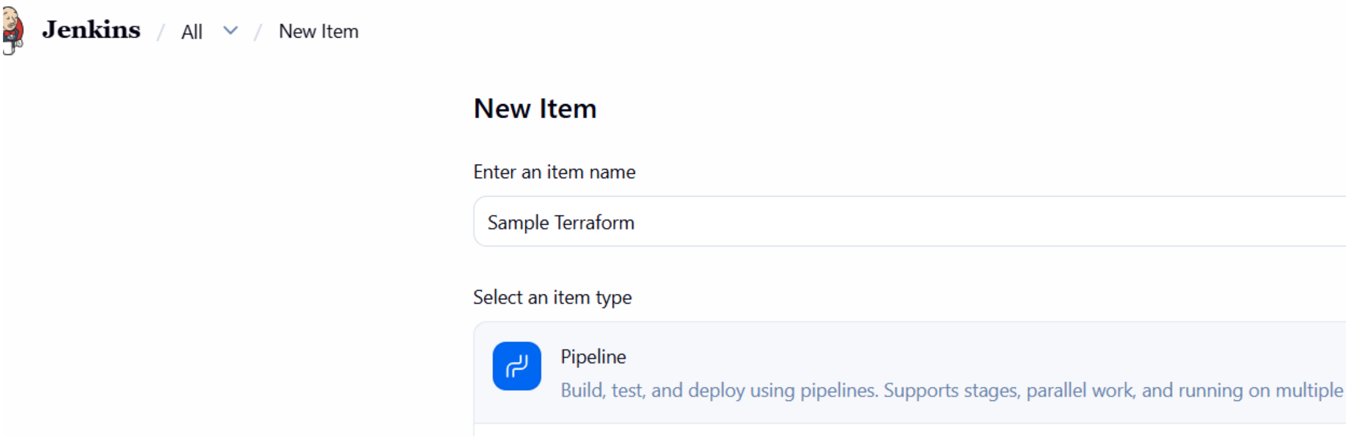
- Open Jenkins in browser and install Terraform plugin



- In Jenkins manage select Tools and add Terraform installations details



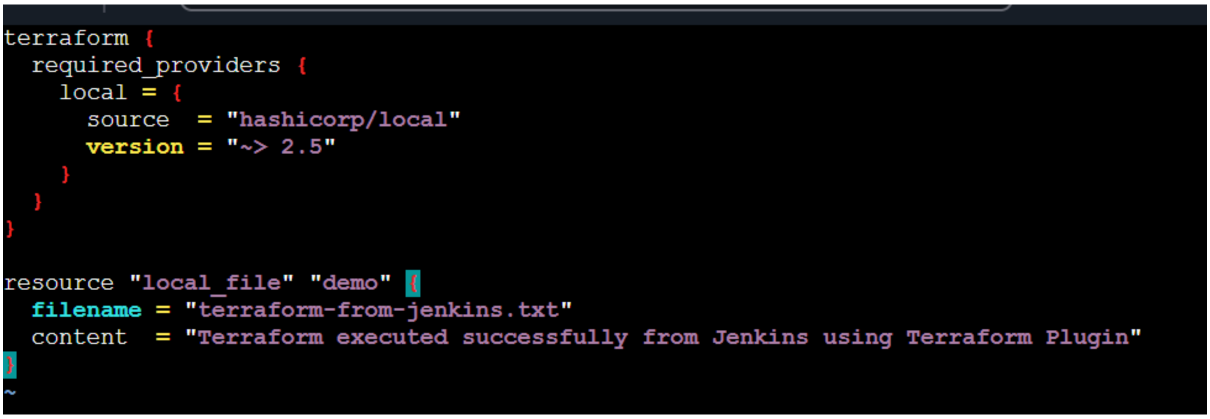
- Now select new job item and select pipeline



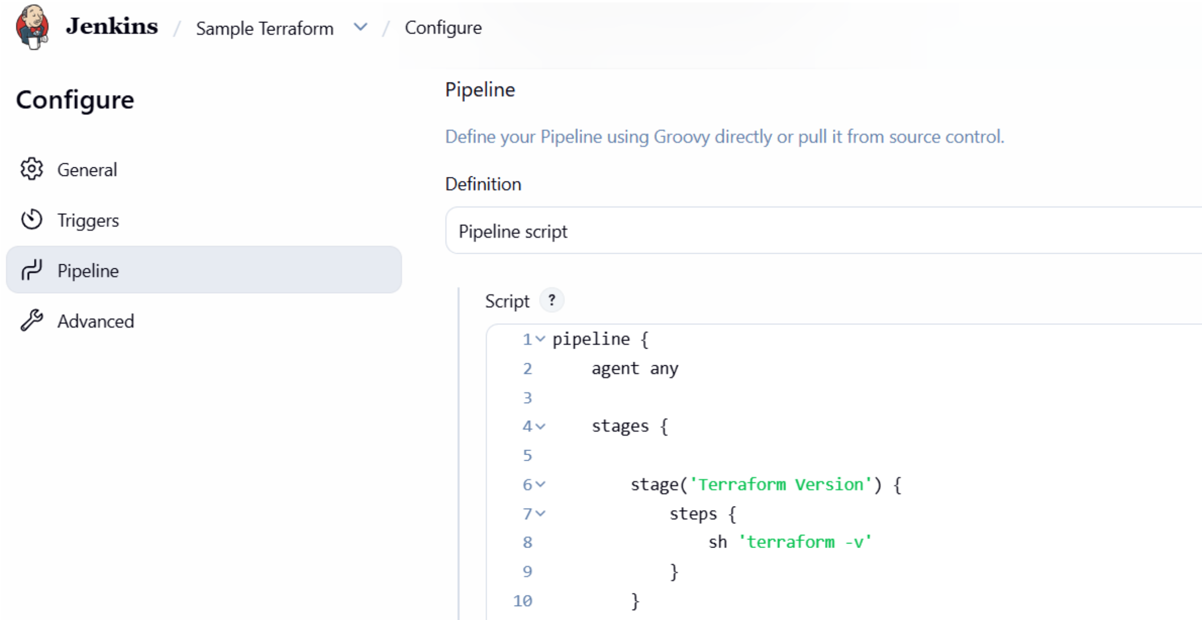
- Now in Jenkins server go to workspace of new created job and there create main.tf

```
root@ip-172-31-22-87:/var/lib/jenkins/workspace# cd ..
root@ip-172-31-22-87:/var/lib/jenkins# cd workspace/Sample\ Terraform
root@ip-172-31-22-87:/var/lib/jenkins/workspace/Sample Terraform# vi main.tf
root@ip-172-31-22-87:/var/lib/jenkins/workspace/Sample Terraform# ls
main.tf
root@ip-172-31-22-87:/var/lib/jenkins/workspace/Sample Terraform#
```

- Now in main.tf file write script and save it



- Now Pipeline and write a declarative script for the new created job



Pipeline Script:

```
pipeline {
  agent any

  stages {

    stage('Terraform Version') {
      steps {
        sh 'terraform -v'
      }
    }

    stage('Terraform Init') {
      steps {
        sh 'terraform init'
      }
```

```
    }
    stage('Terraform Plan') {
      steps {
        sh 'terraform plan'
      }
    }
    stage('Terraform Apply') {
      steps {
        sh 'terraform apply -auto-approve'
      }
    }
  }
}
```

- Successful script

ns / Sample Terraform / #1

```
      ⊠[32m+⊠[0m⊠[0m content_base64sha256 = (known after apply)
      ⊠[32m+⊠[0m⊠[0m content_base64sha512 = (known after apply)
      ⊠[32m+⊠[0m⊠[0m content_md5          = (known after apply)
      ⊠[32m+⊠[0m⊠[0m content_sha1         = (known after apply)
      ⊠[32m+⊠[0m⊠[0m content_sha256       = (known after apply)
      ⊠[32m+⊠[0m⊠[0m content_sha512       = (known after apply)
      ⊠[32m+⊠[0m⊠[0m directory_permission = "0777"
      ⊠[32m+⊠[0m⊠[0m file_permission      = "0777"
      ⊠[32m+⊠[0m⊠[0m filename             = "terraform-from-jenkins.txt"
      ⊠[32m+⊠[0m⊠[0m id                   = (known after apply)
    }

⊠[1mPlan:⊠[0m ⊠[0m1 to add, 0 to change, 0 to destroy.
⊠[0m⊠[1mlocal_file.demo: Creating...⊠[0m⊠[0m
⊠[0m⊠[1mlocal_file.demo: Creation complete after 0s [id=221065dac85086259e58924cd5d
⊠[0m⊠[1m⊠[32m
Apply complete! Resources: 1 added, 0 changed, 0 destroyed.⊠[0m
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
```

- Now check in server successful

```
root@ip-172-31-22-87:/var/lib/jenkins/workspace/Sample Terraform# terraform apply -auto-approve
local_file.demo: Refreshing state... [id=221065dac85086259e58924cd5d80a552d879e04]

No changes. Your infrastructure matches the configuration.

Terraform has compared your real infrastructure against your configuration and found no differences, so no changes are needed.

Apply complete! Resources: 0 added, 0 changed, 0 destroyed.
root@ip-172-31-22-87:/var/lib/jenkins/workspace/Sample Terraform#
```

- To check open Vi terraform-from-jenkins.txt

```
-rwxr-xr-x 1 jenkins jenkins   67 Jun 10 09:07 terraform-from-jenkins.txt
-rw-r--r-- 1 jenkins jenkins 1694 Jun 10 09:09 terraform.tfstate
root@ip-172-31-22-87:/var/lib/jenkins/workspace/Sample Terraform# vi terraform-from-jenkins.txt
root@ip-172-31-22-87:/var/lib/jenkins/workspace/Sample Terraform#
```

Successfully from Jenkins using Terraform plugin

```
Terraform executed successfully from Jenkins using Terraform Plugin
~
~
~
```